What Should Count as a Token?
A Comprehensive Study of Data Representation for Foundation Models

TODO

Abstract

Draft suggestion: Foundation models operate on finite collections of addressable units, yet the choice of these units often receives less study than the architecture that processes them. Tokenization determines unit boundaries, representation scale, retained information, explicit relations, and computational allocation. These choices are especially important in science, where measurement units may differ from the structural or functional units required by a task. We define a tokenizer as a three-module system that constructs reusable parameters, maps an input to a token representation, and maps that representation to a reconstruction. This definition covers algorithmic, neural, and hybrid tokenizers while separating token formation from the downstream foundation model.

Draft suggestion: We study how these methods define tokens for different AI modalities and scientific domain objects, identifying the component tokenizers used to represent individual data sources or objects. We also study system-level tokenization for complex settings that combine multiple modalities, domain objects, spatial or temporal scales, and coupled observations and actions.

Draft suggestion: Across these levels, we examine the trade-offs among coverage, compression, reconstruction, task information, scientific constraints, and model compatibility. We conclude with design principles, evaluation protocols, and a research agenda for adaptive, multiscale, and reusable tokenizers that support future foundation models in prediction, generation, and interaction.

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1 Introduction

Draft suggestion: Extend the related-work coverage by searching for unit formation under segmentation, compression, patching, pooling, quantization, motif discovery, grammar induction, discretization, and state abstraction. Prioritize papers that introduce or directly evaluate tokenization. Retain reuse-only systems when they have broad impact, document a major public model, or supply a comparison that is missing from the review.

1.1 Motivation, Scope, and Contributions

Draft suggestion: State the intended readers, the inclusion rule, and the three contributions in terms that can be checked against the catalog. Keep examples representative rather than exhaustive. Present AI modality and domain object as two views of component tokenizers, then present complex systems as the composition of those components across modalities, objects, scales, or time.

1.2 From Compression to Intelligence

Draft suggestion: Trace the technical line from coding and dictionary compression to subword learning, vector quantization, fixed patches, and learned unit selection. Emphasize which ideas changed tokenizer training, encoding, or decoding. Avoid presenting the history as a sequence of foundation-model architectures.

1.3 Observations and Insights

Draft suggestion: Tokenizer systems are often compared only through their output units. Test whether reported conclusions change after separating tokenizer training, encoding, and decoding. Useful cases include subword vocabularies with different segmentation rules, visual codebooks with different decoders, and molecular vocabularies with incomplete inverse mappings. Keep this observation provisional until each case has direct evidence.

Draft suggestion: Scientific token identity often depends on relations and scale. Compare atoms with motifs, residues with structure units, genes with cell states, and grid cells with adaptive regions. Record when geometry, symmetry, experimental condition, or task changes the useful unit. Include cases where fixed small units remain the stronger choice.

Draft suggestion: Compression, reconstruction, and scientific utility can disagree. Look for tokenizers that reduce sequence length or reconstruction error while losing rare events, valid composition, intervention response, or transfer information. Compare the tokenizer training objective with every later evaluation target before making a general claim.

2 Tokenizer Systems: Definition and Method Taxonomy

Draft suggestion: Use the three-module definition as the common comparison frame for every method family. Classify only the operations that determine token units, assignments, or reconstruction. Keep embedding layers, foundation-model backbones, and task heads outside the tokenizer unless they change one of these operations.

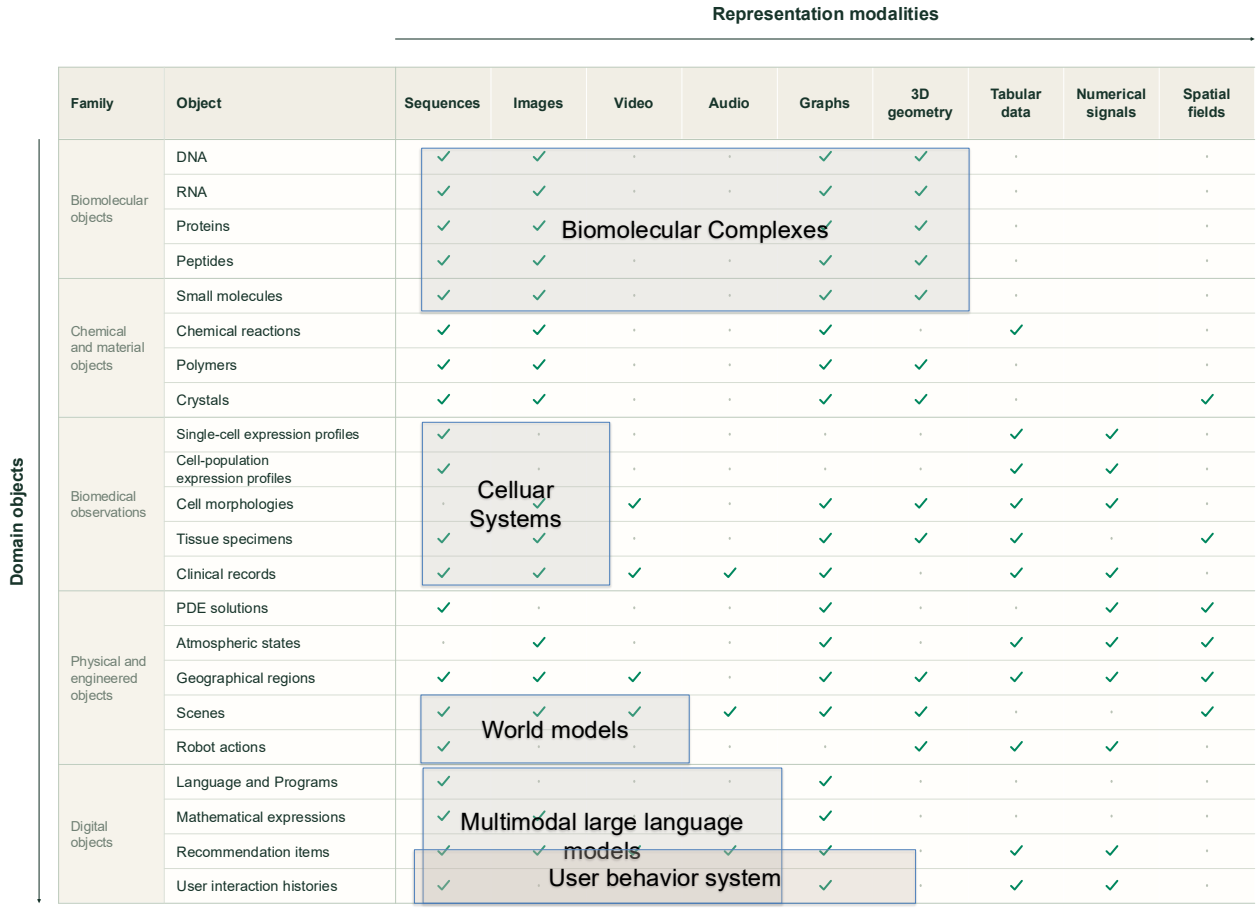


Figure 1: TODO: Overview of the work.

2.1 A Tokenizer as a Three-Module System

Draft suggestion: Keep the definition independent of modality and downstream architecture. Explain the roles of `train`, `encode`, and `decode` through BPE, VQ-VAE, and fixed image patches. State the reconstruction and equivalence rule for lossy or structured data, then use the definition to separate full tokenizers from methods that specify only one module.

Let $\mathcal{X}$ denote an input space and $\mathcal{Z}$ a token space. We define a tokenizer system as

$$\mathcal{T} = \big(\text{train}, \text{encode}, \text{decode}\big), \tag{1}$$

with operations

$$\theta = \text{train}(\mathcal{D}), \qquad Z = \text{encode}_\theta(x), \qquad \widehat{x} = \text{decode}_\theta(Z). \tag{2}$$

Here, $\mathcal{D}$ is the data used to construct or fit the tokenizer, $x \in \mathcal{X}$ is an input, $Z \in \mathcal{Z}$ is its token representation, and $\widehat{x} \in \mathcal{X}$ is the reconstructed input. We use θ for all reusable quantities returned by training. These quantities may include a vocabulary, merge table, or codebook. They may also include deterministic rules or neural parameters. A predefined tokenizer is the special case $\text{train}(\mathcal{D}) = \theta_0$ for every $\mathcal{D}$. For a structured input, the encoded representation can be written as

$$Z = \big((z_i)_{i=1}^{n}, R\big), \tag{3}$$

where z_i is one token and R records any relations required to interpret the token collection. Sequence order can define R implicitly. Graph edges, spatial coordinates, or hierarchical links may define it explicitly. A unit qualifies as a token when a later model can process it separately for masking, prediction, or routing. The decoder recovers what information the tokenizer preserves. Exact recovery requires

$$\text{decode}_\theta\big(\text{encode}_\theta(x)\big) = x. \tag{4}$$

Some domains have several records for the same object. In such cases, recovery may be defined up to a stated equivalence, such as graph isomorphism, geometric symmetry, or text normalization. Continuous data usually require an approximation for reconstruction rather than equality. Therefore, the reconstruction target and its comparison rule are part of the tokenizer definition.

We categorize tokenizers into three classes based on how their training algorithms are defined: (1) algorithmic tokenizers, (2) neural tokenizers, and (3) hybrid tokenizers. Algorithmic tokenizers use non-neural training algorithms, which can be either predefined or data-driven, whereas neural tokenizers rely on generative models and quantization techniques.

Byte pair encoding (BPE) is an example of an algorithmic tokenizer. Its training function constructs a vocabulary and an ordered merge table from corpus statistics. The encoding function applies the learned merges to a new sequence, while the decoding function concatenates the resulting symbols.

Another example of an algorithmic tokenizer is image patching, as used in the Vision Transformer (ViT) [1]. In this predefined tokenizer, θ_0 specifies the patch size and ordering. In contrast, vector-quantized variational autoencoders (VQ-VAEs) are examples of neural tokenizers [2] and have been adapted to continuous and non-discrete data. In the following subsections, we define these three classes in detail.

2.2 Algorithmic Tokenizers

Draft suggestion: Develop this subsection around the three sources of unit formation rather than a long list of algorithms. For each family, compare what `train` estimates, how `encode` resolves competing segmentations, what `decode` recovers, and which choices introduce randomness. Select a small number of examples from text, signals, graphs, and scientific structures.

2.3 Neural and Hybrid Tokenizers

Draft suggestion: Ask section authors to identify the last operation that changes token identity or membership. Use that boundary to distinguish the tokenizer from embedding layers, backbones, task heads, and generators. Compare neural systems by their learned token operation and compare hybrid systems by how the algorithmic and neural stages are composed.

A vector-quantized variational autoencoder (VQ-VAE) gives a neural example [2]. The training function jointly learns a neural network encoder f_ϕ, a codebook $C = \{c_j\}_{j=1}^{K}$, and a neural network decoder g_ψ. For each local feature, the encoding function assigns

$$k_i = \arg\min_{1 \le j \le K} \|f_\phi(x)_i - c_j\|_2, \qquad \widehat{x} = g_\psi(c_{k_1}, \ldots, c_{k_n}). \tag{5}$$

The tokens are the code indices $(k_i)_{i=1}^{n}$, and decoding is generally approximate. The codebook is part of θ.

3 Data Representation and Tokenization across AI Modalities

Draft suggestion: Write this section for machine learning researchers and organize it by representation form. Use general AI tasks and models to compare token units, relations, training objectives, decoding targets, token rates, and computational cost. Mention scientific examples such as SMILES strings, protein sequences, molecular graphs, or physical trajectories only briefly, then refer readers to Section 4 for their object-specific meaning and tests. This section should explain transferable tokenization methods rather than repeat the scientific-object review.

3.1 Sequences

Draft suggestion: Representative work: Sennrich et al. [3]. Treat a sequence as an ordered representation and compare characters, bytes, subwords, words, and syntax-aware units. Use text and code as the main AI cases. Discuss coverage, sequence length, reversibility, multilingual behavior, and syntactic structure. Mention SMILES, nucleic-acid sequences, and protein sequences only as short transfer examples, then refer readers to Section 4.

3.2 Images

Draft suggestion: Representative work: Dosovitskiy et al. [1] and Van Den Oord et al. [2]. Compare pixels, fixed patches, regions, object-level units, and latent visual codes for recognition and generation. Separate token boundaries from token values and compare reconstruction, token rate, spatial resolution, and compatibility with vision backbones. Treat scientific grids as images only when their row-and-column layout defines the model input; discuss their scientific meaning in Section 4.

3.3 Video

Draft suggestion: Use video tokenization work that studies frames, tubelets, motion-aware units, and discrete visual codes. Explain how temporal continuity changes boundary formation, token rate, and decoding compared with images. Discuss scientific video only briefly here and refer object-specific questions about calibrated intensity, event timing, and identity across frames to Section 4.

3.4 Audio

Draft suggestion: Representative work: Baevski et al. [4]. Compare waveform samples, frames, learned speech units, and neural codec codes while separating linguistic content from acoustic information. Cover speech, music, and sound events as distinct uses of the same representation modality. Keep generic sampled measurements in the time-series and signals subsection.

3.5 Graphs

Draft suggestion: Representative work: Galkin et al. [5]. Compare nodes, edges, paths, neighborhoods, motifs, and learned graph units. State whether adjacency belongs to the token representation or remains external relation data. Discuss permutation, structural roles, graph reconstruction, and vocabulary transfer. Use molecular, crystal, protein, and cell graphs only as short examples before referring readers to Section 4.

3.6 3D Geometry

Draft suggestion: Representative work: Yu et al. [6]. Compare points, voxels, mesh elements, local geometric patches, and object-level units when coordinates, shape, or spatial neighborhoods determine token identity. Discuss sampling density, rigid transformations, local frames, and geometric reconstruction. Treat a mesh as a graph when combinatorial adjacency defines the units, and treat it as 3D geometry when coordinates and shape define them.

3.7 Tables

Draft suggestion: Compare cells, columns, rows, records, and learned groups as candidate table tokens. Explain how schema, data type, missing values, and row or column order affect token identity. Treat sets as order-free inputs within this discussion when their elements share the same representation rules. Refer gene-expression matrices, clinical measurements, and interaction histories to their domain-object subsections.

3.8 Time Series and Signals

Draft suggestion: Representative work: Lin et al. [7]. Compare samples, windows, events, transformed coefficients, and quantized measurements for forecasting, anomaly detection, and numerical sequence modeling. Distinguish time-indexed measurements from video, where spatial appearance remains central, and from audio, where waveform and acoustic structure define the task. Mention molecular trajectories, physical simulations, and Earth dynamics only to show method transfer, then refer their scientific tests to Section 4.

Table 1: Representation forms and their main tokenization questions.

Modality	Common token choices	Main comparison question
Text, code, and symbolic sequences	Words, bytes, subwords, syntax units	How do coverage and structural alignment trade off?
Images and video	Patches, regions, objects, latent codes	Which boundaries retain objects and short events?
Audio and speech	Frames, spectral bins, codec codes	Which information needs a separate rate or stream?
Three dimensional geometry	Points, voxels, mesh elements, regions	Which units remain stable under sampling and transformation?
Graphs, tables, and sets	Nodes, subgraphs, cells, records, groups	Which roles and relations define token identity?
Time series and numerical signals	Points, windows, events, coefficients, bins, peaks	Which units retain changes across sampling rates and calibrated axes?

4 Data Representation and Tokenization by Domain Object

Draft suggestion: Organize this section by what is represented and use the modality methods in Section 3 as technical background. Use the six families below only to structure the review. Place each concrete object in a subsubsection. Avoid repeating BPE, patching, graph pooling, vector quantization, or signal segmentation. Explain instead how object identity, accepted equivalences, known relations, measurement conditions, and validation rules change the tokenization problem. Refer systems that combine several objects or modalities to Section 5.

4.1 Language and Formal Objects

Draft suggestion: Use this family to compare tokenization for natural language, source code, and mathematical expressions without treating them as one representation. The sequence methods in Section 3 provide the shared background. Focus here on the linguistic, syntactic, and formal constraints that determine valid boundaries and decoding.

4.1.1 Natural Language

Draft suggestion: Representative work: Sennrich et al. [3]. Discuss words, characters, bytes, and learned subwords through input coverage, linguistic boundaries, multilingual transfer, normalization, and exact decoding. Separate the tokenizer from the language model that consumes its output.

4.1.2 Code and Mathematics

Draft suggestion: Compare lexical tokens, learned subwords, syntax-tree units, and serialized mathematical expressions. Ask how a tokenizer handles identifiers, whitespace, formal grammar, variable names, and equivalent expressions. Use sequence and graph tokenization from Section 3 as background, then focus on syntactic validity and executable or symbolic equivalence.

4.2 Music and Sound Events

Draft suggestion: Keep this family separate from natural language because musical structure and non-speech events use different units and decoding targets. Build on the audio methods in Section 3, then compare event tokens, note-like symbols, acoustic codec codes, and their temporal hierarchy.

4.2.1 Music and Sound Events

Draft suggestion: Discuss symbolic music, acoustic music, and non-speech events through timing, pitch or event identity, polyphony, hierarchy, and waveform reconstruction. State whether tokens represent semantic events, acoustic segments, or codec codes, because these choices support different prediction and generation tasks.

4.3 Molecular and Material Objects

Draft suggestion: This family groups objects whose token definitions depend on chemical identity, bonding, geometry, composition, or periodicity. Keep the five object subsubsections distinct. Compare them through valid transformations, reconstruction targets, scale, and physical constraints rather than through one shared chemical vocabulary.

4.3.1 Nucleic Acids

Draft suggestion: Representative work: Dalla-Torre et al. [8]. Sequence methods overlap with Section 3; summarize them briefly and focus here on bases, codons, regulatory regions, reverse complements, reading frames, mutations, and transfer across genomes. Move joint nucleic-acid and protein systems to Section 5.

4.3.2 Proteins and Peptides

Draft suggestion: Representative work: Rives et al. [9]. This object overlaps with sequences, graphs, 3D geometry, and time series in Section 3. Focus on residues, motifs, domains, structural units, conformational states, noncanonical residues, and biological function. Include peptides when residue sequence, conformation, or biological function defines the problem. Discuss protein complexes and multiscale biology in Section 5.

4.3.3 Small Molecules and Reactions

Draft suggestion: Representative work: Jin et al. [10]. Sequences, graphs, 3D geometry, spectra, and trajectories overlap with Section 3; focus here on molecular equivalence, stereochemistry, valence, fragments, reaction centers, chemical validity, and inverse reconstruction. Treat reactions as transformations among molecular objects. Move biomolecular assemblies with proteins or nucleic acids to Section 5.

4.3.4 Synthetic Polymers

Draft suggestion: Representative work: Zhu et al. [11]. Discuss monomers, repeat units, chains, branches, conformations, and learned motifs. Check whether the tokenizer treats repeat-unit equivalence, sequence disorder, molecular-weight distribution, chain architecture, and processing history as part of the representation or as external conditions.

4.3.5 Crystals

Draft suggestion: Compare atom-level, graph-based, geometric, and sequence representations of periodic structures. Focus on unit-cell choice, periodic boundary conditions, atom permutation, symmetry, defects, composition, and reconstruction of an equivalent crystal. Keep diffraction as a measurement linked to the crystal rather than as a separate domain object.

4.4 Biomedical Observations

Draft suggestion: This family moves from molecular measurements to spatial morphology and clinical context. Keep the three object subsubsections distinct because their samples, relations, missing-data processes, and validation protocols differ. Use the modality section for image, table, graph, sequence, and signal methods.

4.4.1 Gene Expression Profiles

Draft suggestion: Representative work: Cui et al. [12]. Compare genes, expression values, ranked gene lists, expression bins, cells, and learned cell states. Discuss assay conditions, sparsity, batch effects, perturbations, and transfer across tissues or studies. Distinguish gene tokens from cell-level sample tokens.

4.4.2 Cell and Tissue Morphology

Draft suggestion: Discuss microscopy and digital pathology through pixels, patches, segmented cells, subcellular structures, tissue regions, and spatial neighborhoods. Compare token scale against rare-cell retention, tissue boundaries, spatial context, and cross-instrument transfer. Refer multimodal or perturbation systems that combine morphology with molecular measurements to Section 5.

4.4.3 Clinical Records and Measurements

Draft suggestion: Compare tokenization of clinical text, codes, laboratory measurements, physiological signals, medical images, and longitudinal events. Focus on missingness, irregular time, coding systems, patient context, privacy constraints, and validation across clinical sites. Keep modality-specific mechanics in Section 3.

4.5 Physical and Robotic Objects

Draft suggestion: This family covers physical variables, represented environments, and control outputs. Keep fields, scenes, and actions separate because they differ in their coordinate systems, decoding targets, and evaluation. Discuss a complete action-conditioned model of observations and transitions in Section 5.

4.5.1 Physical Fields

Draft suggestion: Representative work: Wu et al. [13]. Build on image, graph, geometry, and time-series methods from Section 3. Focus on physical variables, units, boundary conditions, conservation laws, discretization error, numerical stability, and transfer across grids or meshes.

4.5.2 Scenes and Geographical Regions

Draft suggestion: Representative work: Lam et al. [14]. Compare scene regions, objects, map cells, geographic regions, and geospatial graph nodes. Distinguish generic 3D geometry from location-specific coordinates, map projections, vertical levels, observation networks, and transfer across locations or seasons. Move joint state, action, and transition tokenization to Section 5.

4.5.3 Robot Actions

Draft suggestion: Representative work: Pertsch et al. [15]. Compare scalar controls, transformed coefficients, discrete actions, action chunks, and reusable action segments. Evaluate action reconstruction, control error, temporal abstraction, and closed-loop behavior. Exclude systems whose main contribution is joint observation, latent-state, action, and transition tokenization; discuss them as world models in Section 5.

4.6 Recommendation Data

Draft suggestion: Use this family for tokenization that represents recommendable items and observed interaction histories. Keep its vocabulary aligned with recommender-system research. Treat recommendation as the application context and place the concrete represented objects in the subsubsection below.

4.6.1 Items and Interaction Histories

Draft suggestion: Begin with generative retrieval with semantic identifiers. Compare item identifiers, semantic identifiers, item sequences, sessions, and temporal histories. Ask whether the tokenizer preserves item identity, item coverage, interaction order, new-item handling, and decoding to valid items.

5 Tokenization for Complex Systems

Draft suggestion: Treat Sections 3 and 4 as the component library for this section. Include a system here only when its tokenizer combines components across modalities, domain objects, biological scales, or temporally coupled observations and actions. Multiple representation labels alone are insufficient. Do not repeat the construction of each component tokenizer. Study instead how components are aligned, kept separate, assigned token rates, related across scales or time, decoded jointly, and tested as one system.

5.1 Multimodal Language Models

Draft suggestion: Representative work: Lu et al. [16]. This subsection builds directly on the modality-specific tokenizers in Section 3. Compare separate, aligned, and shared token spaces through cross-modal correspondence, token-rate balance, attention allocation, and decoding support. Focus on models that combine language with images, video, audio, or other represented inputs and outputs. Distinguish this setting from biomolecular complexes, cellular perturbation systems, and world models, whose system structure follows scientific components or temporal transitions.

5.2 Biomolecular Complexes

Draft suggestion: Representative work: Abramson et al. [17]. Build on the nucleic-acid, protein, peptide, and small-molecule tokenizers in Section 4. Study how one system preserves component identity while representing cross-component contacts, stoichiometry, geometry, ions, ligands, and missing structures. Do not repeat each component tokenizer. Compare instead how the system aligns, relates, and jointly decodes several biomolecular objects.

5.3 Cellular Perturbation Systems

Draft suggestion: Representative starting point: Cui et al. [12]. Build on gene expression profiles and cell morphology in Section 4. Study how a tokenizer relates molecular interventions, cell state, time, dose, morphology, and measured response. Distinguish this subsection from biomolecular complexes, which model interacting components within an assembly, and from world models, which center on action-conditioned transitions in an environment. Compare how systems represent perturbations, preserve cell identity, align assays, and decode predicted responses.

5.4 World Models

Draft suggestion: Representative work: Bruce et al. [18]. Study systems that jointly tokenize observations, latent states, actions, and transitions. Distinguish them from Section 4, which treats scenes and robot actions as separate domain objects, and from multimodal language models, which need not model action-conditioned change. Include action-only methods such as FAST in the domain-object section and cite them here only when comparing the action component of a complete world model. Evaluate temporal identity, rollout stability, controllability, and closed-loop behavior at the system level.

6 Tokenizer Design and Evaluation

Draft suggestion: Connect every proposed property to a design choice and a measurable test. Separate properties of the tokenizer from outcomes of the downstream foundation model. Use the companion catalog to select representative systems, and move full record-level statistics to supplementary material after human review.

6.1 Core Properties of a Tokenizer System

Draft suggestion: Define a short set of properties that applies across modalities: input coverage, stable encoding, declared reconstruction, token rate, and compatibility with the consuming model. State which properties are required and which depend on task or data. Treat exact reconstruction as a requirement only when the declared representation is reversible.
https://arxiv.org/pdf/2511.11758#page=7.48: GeoBPE compression, reconstruction, generalization, interpretability

6.2 Task and Scientific Design Principles

Draft suggestion: Organize this subsection around prediction, generation, unified prediction and generation, and scientific data. For prediction, test discriminative information, generalization, and multiple scales. For generation, test reconstructability, compositional control, validity, and token rate. For science, test units, relations, symmetry, causality, measurement conditions, and response under intervention.

6.3 Benchmarks and Comparisons

Draft suggestion: Evaluate three linked outcomes: representation and system cost, reconstruction and scientific validity, and downstream utility. Match data, backbone, optimization, and compute where the claim concerns tokenization. Report tokenizer-training data separately, include repeated runs for learned tokenizers, and test rare cases, valid transformations, and realistic distribution shifts.

7 Future Directions

Draft suggestion: Treat foundation-model construction as the end goal while separating tokenizer-specific evidence from model-dependent evidence. Organize the research agenda around three questions: how tokenizers should be evaluated with and without foundation models, what foundation models require from tokenizers, and how the two systems should be designed together. For each direction, state a testable claim, a controlled comparison, and the conditions under which a simpler tokenizer remains preferable.

7.1 Evaluating Tokenizers with and without Foundation Models

Draft suggestion: Representative foundations: rate–distortion theory [19], VQ-VAE [2], and the scaling analysis of Byte Latent Transformer [20]. Organize evaluation around compression, intelligence, and their joint scaling. Without a foundation model, measure input coverage, encoding stability, token rate, vocabulary or codebook use, reconstruction under the declared equivalence, and the cost of training, encoding, and decoding. Report lossless rate when exact recovery is required, and report distortion together with structural or scientific validity for lossy tokenizers. With a foundation model, test whether the retained information supports prediction, generation, simulation, and control under matched data, backbone, optimization, parameter count, context budget, and total compute. Connect the two levels through rate–distortion–utility curves across model and data scales. Use raw bytes, pixels, residues, atoms, or observations as common denominators because token counts vary across tokenizers. Include reuse as a model-conditioned test: compatibility asks whether models can consume the same token meanings; transferability asks whether a tokenizer remains useful on new data or tasks; replaceability asks how much adaptation is required to substitute one tokenizer for another. Separate tokenizer-training and retokenization costs from foundation-model training, and do not attribute a full-system gain to tokenization without a controlled comparison.

7.2 Designing Tokenizers for Foundation Models

Draft suggestion: Representative work: Lu et al. [16] and Pertsch et al. [15]. Derive tokenizer requirements from the foundation model that consumes its output. Ask how the learning objective, attention pattern, context length, positional representation, output decoder, and deployment budget constrain unit boundaries, token rate, vocabulary or codebook size, and relation encoding. Prediction requires discriminative information and generalization across valid changes. Generation requires sufficient information for reconstruction, valid composition, and controllable output length. Simulation and action further require temporal identity, causal order, and stable closed-loop decoding. Study whether one tokenizer can support these uses or whether shared primitive units should feed task-specific compositions. Require deterministic encoding when models or checkpoints must share token identifiers. Test whether the same tokenizer can serve different foundation-model sizes and architectures, and state the adaptation required when it cannot. For scientific foundation models, add object identity, symmetry, conservation, measurement conditions, and scale as explicit design constraints rather than inferring them from downstream accuracy.

7.3 Co-designing Tokenizers and Foundation Models

Draft suggestion: Representative work: Ryoo et al. [21], Pagnoni et al. [20], and Lu et al. [16]. Reverse the usual design question and ask how the foundation model should change when the tokenizer returns variable-length, overlapping, hierarchical, graph-structured, continuous, or stochastic units. Study positional and relation encodings that respect token geometry and hierarchy; attention and routing that allocate computation according to token scale, uncertainty, or information content; and training objectives that evaluate predictions in the original data space as well as the token space. A model should retain token boundaries, parent–child relations, and decoding uncertainty when these quantities affect the task. Joint training should control codebook collapse, unstable boundaries, and shifts in token meaning, while preserving a clear encoding and decoding procedure. Compare fixed-token backbones with architectures designed for the same tokenizer under matched total compute. Extend checkpoint reuse through token mappings, embedding adaptation, or partial retraining, and report the performance recovery cost. Treat algorithmic tokenizers as valid co-design choices: co-design concerns coordinated assumptions between modules and does not require an end-to-end neural tokenizer.

8 Conclusion

Draft suggestion: End the final manuscript with the strongest claim supported by the completed evidence, followed by one clear limitation and one open research question. Remove claims that depend only on unmatched full-model comparisons or unverified tokenizer details.

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